

Functional Annotation Report: PP_2325 (Q88KG5) — Ketopantoate Reductase of *Pseudomonas putida* KT2440

Summary

PP_2325 (UniProt Q88KG5) of *Pseudomonas putida* KT2440 encodes ketopantoate reductase (KPR), also known as 2-dehydropantoate 2-reductase (PanE/ApbA; EC 1.1.1.169; KEGG orthology K00077). This soluble, cytoplasmic oxidoreductase catalyzes the NADPH-dependent, stereospecific reduction of 2-dehydropantoate (ketopantoate) to (R)-pantoate. This reaction is the committed reductive step in the pantoate branch of pantothenate (vitamin B5) biosynthesis, which in turn feeds the biosynthesis of coenzyme A (CoA) and acyl carrier protein (ACP) — cofactors central to all fatty-acid, polyketide, and central-carbon metabolism. The substrate is specifically ketopantoate and the obligate hydride donor is NADPH.

The identity of PP_2325 as a ketopantoate reductase is strongly supported by convergent lines of evidence. UniProt and KEGG independently annotate the 315-residue protein as 2-dehydropantoate 2-reductase (EC 1.1.1.169, K00077), placing it in pathway ppu00770 (Pantothenate and CoA biosynthesis). The protein carries the diagnostic InterPro/Pfam signatures of the family — ApbA (PF02558; the N-terminal NAD(P)-binding Rossmann domain) and ApbA_C (PF08546; the C-terminal catalytic domain) — and directly retains the N-terminal Rossmann glycine-rich dinucleotide-binding motif (GSGAIG at residues 12–17). Function is transferred with high confidence from biochemically and structurally characterized bacterial orthologs (chiefly *Escherichia coli* PanE), for which the reaction, cofactor specificity, kinetic mechanism, stereochemistry, and catalytic residues have all been experimentally established.

Because the ketopantoate reductase family is intrinsically divergent at the sequence level (~20–29% pairwise identity between validated homologs), residue-level catalytic conservation in PP_2325 is best inferred through profile-HMM/InterPro methods rather than naive pairwise alignment. PP_2325 shows ~27% identity to *E. coli* PanE, which is entirely within the normal range for this family. The pantothenate pathway is otherwise complete in *P. putida* (panBC at

PP_4699/PP_4700), and an IlvC ortholog (PP_4678) plus two additional KPR-family paralogs (PP_1351, PP_2998) may provide partial functional redundancy. Among these, PP_2325 is the canonically curated PanE/ApbA ketopantoate reductase.

Key Findings

Finding 1 — PP_2325 is ketopantoate reductase (EC 1.1.1.169), catalyzing the NADPH-dependent reduction of 2-dehydropantoate to (R)-pantoate

UniProt Q88KG5 annotates PP_2325 as 2-dehydropantoate 2-reductase (EC 1.1.1.169) with the catalytic activity:



The 315-amino-acid protein belongs to the ketopantoate reductase (ApbA/PanE) family, carrying InterPro signatures IPR003710 (ApbA) and IPR013752 (KPA_reductase). Direct sequence inspection confirms an N-terminal Rossmann-fold dinucleotide-binding glycine-rich motif (GSGAIG at residues 12–17), which matches the canonical G-x-G-x-x-[GA] fingerprint for NADP(H) binding.

This assignment is anchored in primary biochemical literature on the enzyme family. As stated in the structural characterization of *E. coli* KPR, "*The NADPH-dependent reduction of ketopantoate to pantoate, catalyzed by ketopantoate reductase (KPR; EC 1.1.1.169), is essential for the biosynthesis of pantothenate (vitamin B5)*" [PMID: 15966718](#). This is independently reaffirmed in the ternary-complex study: "*Ketopantoate reductase (KPR, EC 1.1.1.169) catalyzes the NADPH-dependent reduction of ketopantoate to pantoate, an essential step for the biosynthesis of pantothenate (vitamin B5)*" [PMID: 17229734](#). Both establish the reaction, the cofactor (NADPH), the EC number, and the pathway role that match the UniProt annotation of PP_2325.

Finding 2 — KPR is a two-domain Rossmann-fold enzyme using conserved Lys, Asn, and Glu residues with a substrate-induced conformational change

Crystallographic and kinetic work on *E. coli* KPR — a monomeric homolog — reveals a two-domain architecture: NADP(H) is bound in a cleft between an N-terminal Rossmann-fold domain and a C-terminal α -helical domain. As reported, *"The cofactor is bound in the active site cleft between the N-terminal Rossmann-fold domain and the C-terminal alpha-helical domain"* PMID: 15966718.

Site-directed mutagenesis coupled with steady-state kinetics has dissected the catalytic machinery: *"The steady-state kinetics of active site mutants R31A, K72A, N98A, K176A, S244A, and E256A implicate Asn98 as well as Lys176 and Glu256 in the catalytic mechanism"* PMID: 17229734. In this scheme, Asn98, Lys176, and Glu256 form the catalytic core; Arg31 and Lys72 bind the cofactor; and Ser244 contacts the substrate. Catalysis involves a hinge-bending domain closure that repositions the essential Lys176 to hydrogen-bond the C2 hydroxyl of the pantoate product. PP_2325 conserves the corresponding N-terminal Rossmann motif (GSGAIG) and is assigned to the same family, so it is expected to employ the same two-domain catalytic strategy.

Finding 3 — KPR is a monomeric ~34 kDa enzyme with an ordered sequential mechanism and stereospecific pro-S hydride transfer to C2

Detailed kinetic and NMR analysis of the *E. coli* panE-encoded enzyme established the reaction mechanism. The enzyme is a monomeric ~34 kDa protein with $K_M(\text{NADPH}) \approx 20 \mu\text{M}$, $K_M(\text{ketopantoate}) \approx 60 \mu\text{M}$, and $k_{\text{cat}} \approx 40 \text{ s}^{-1}$. Steady-state and product-inhibition patterns fit an ordered sequential mechanism: *"The steady-state initial velocity and product inhibition patterns are consistent with an ordered sequential kinetic mechanism in which NADPH binding is followed by ketopantoate binding, and pantoate release precedes NADP(+) release"* PMID: 10736170.

The chemistry is stereospecific: *"The stereospecific transfer of the pro-S hydrogen atom of NADPH to the C-2 position of ketopantoate was demonstrated by (1)H NMR spectroscopy"* PMID: 10736170, which yields D-(–)-(R)-pantoate. A single general acid/base ($\text{pK} \approx 7.8\text{--}8.4$) operates during turnover, and hydride transfer is not fully rate-limiting. The monomeric two-domain fold is corroborated by the 1.7 Å crystal structure: *"The enzyme is monomeric and has two domains separated by a cleft. The N-terminal domain has an alphabeta-fold of the Rossmann type"* PMID: 11724562. PP_2325 (315 aa, ~34 kDa) is the *P. putida* ortholog of this enzyme and is expected to share these properties.

Finding 4 — PP_2325 performs the final step of the pantoate branch of pantothenate/CoA biosynthesis in the cytoplasm; it belongs to the 6-PGDH C-terminal-domain-like superfamily

The bacterial pantothenate biosynthetic pathway comprises four enzymes. As summarized: *"the pantothenate biosynthetic pathway is well-established in bacteria, comprising four enzymic reactions catalysed by ketopantoate hydroxymethyltransferase (KPHMT), L-aspartate- α -decarboxylase (ADC), pantothenate synthetase (PS) and ketopantoate reductase (KPR) encoded by panB, panD, panC and panE genes, respectively"* [PMID: 18726075](#).

UniProt places PP_2325 at *"(R)-pantothenate biosynthesis; (R)-pantoate from 3-methyl-2-oxobutanoate: step 2/2"*. In sequence, PanB (KPHMT) converts 3-methyl-2-oxobutanoate (ketoisovalerate) into ketopantoate; KPR (PP_2325) then reduces ketopantoate to (R)-pantoate; PanC (pantothenate synthetase) condenses (R)-pantoate with β -alanine to form pantothenate, the direct precursor of CoA and ACP.

Structurally, KPR belongs to a larger superfamily: *"KPR is shown to be a member of the 6-phosphogluconate dehydrogenase C-terminal domain-like superfamily"* [PMID: 11724562](#), which matches PP_2325's InterPro assignments IPR008927 and IPR013328. The gene has a documented dual identity: in *Salmonella*, *"apbA and panE are allelic and map to 10 min on both the S. typhimurium and E. coli chromosomes"* [PMID: 9721324](#), confirming that the ApbA and PanE names denote the same ketopantoate reductase gene product. In *E. coli*, a second, moonlighting KPR activity is provided by ilvC (ketol-acid reductoisomerase).

Finding 5 — PP_2325 is a divergent but bona fide KPR ortholog; pantothenate biosynthesis is conditionally essential in *P. putida* KT2440 on minimal medium

Global pairwise alignment (BLOSUM62) of PP_2325 (315 aa) against the experimentally characterized *E. coli* PanE (303 aa) yields 27.0% identity over 293 aligned positions. This is squarely within the documented low-similarity range for the KPR family. As directly stated: *"The sequence similarity among putative KPRs is limited and KPR itself belongs to a large superfamily of 6-phosphogluconate dehydrogenases"* [PMID: 20876192](#). Despite this divergence, PP_2325 retains the diagnostic N-terminal Rossmann NADP-binding motif and the KPR-specific InterPro signatures (ApbA IPR003710, KPA_reductase IPR013752).

Physiologically, a genome-wide mini-Tn5 mutant screen of *P. putida* KT2440 on glucose minimal medium recovered vitamin/amino-acid biosynthesis auxotrophs: *"Among the set of identified conditionally essential genes were a number of genes involved in the biosynthesis of*

certain amino acids and vitamins" PMID: 20158506. Because de novo pantothenate/CoA production requires KPR activity, the enzyme is expected to be required for growth unless pantothenate is supplied. As in *E. coli*, a secondary KPR activity from ketol-acid reductoisomerase (IlvC) may partly mask the phenotype of a single panE knockout.

Finding 6 — Per-residue catalytic-site conservation cannot be verified by naive pairwise alignment; HMM/InterPro annotation is the appropriate method for this divergent family

Attempts to map the experimentally defined *E. coli* PanE catalytic/binding residues (Arg31, Lys72, Asn98, Lys176, Ser244, Glu256, plus substrate-contacting Asn180/194/241) directly onto PP_2325 by hand-rolled global alignment were inconclusive. A linear-gap Needleman–Wunsch alignment gave 27% identity, whereas an affine-gap (Gotoh) alignment gave only 19% identity with the catalytic columns clearly misregistered (e.g., *E. coli* K72 aligning to a non-equivalent PP position 37). This instability reflects the intrinsically low sequence similarity of the family rather than the absence of the catalytic residues — the same limitation noted in the family: *"The sequence similarity among putative KPRs is limited and KPR itself belongs to a large superfamily of 6-phosphogluconate dehydrogenases"* PMID: 20876192.

Reliable residue-level equivalence therefore requires profile-HMM or structure-based alignment. PP_2325 is positively assigned KPR-specific InterPro/Pfam signatures (ApbA IPR003710, KPA_reductase IPR013752) that are built from curated multiple alignments encoding the conserved catalytic positions, and it directly retains the N-terminal Rossmann NADP-binding motif (GSGAIG, residues 12–17) confirmed by sequence inspection. This finding is methodologically important: it explains why the annotation rests on HMM/domain evidence rather than a simple pairwise residue match.

Finding 7 — KEGG independently assigns PP_2325 to K00077 (EC 1.1.1.169); Pfam ApbA + ApbA_C confirm both functional domains; the gene is not in a pan operon

KEGG (entry `ppu:PP_2325`) independently annotates the gene as 2-dehydropantoate 2-reductase, KEGG Orthology K00077, EC 1.1.1.169, mapped to pathways ppu00770 (Pantothenate and CoA biosynthesis) and ppu01240 (Biosynthesis of cofactors), and classified as an oxidoreductase acting on CH-OH groups with NAD(P)⁺ as acceptor. KEGG lists Pfam motifs

ApbA (PF02558; N-terminal NAD(P)-binding/Rossmann domain) and ApbA_C (PF08546; C-terminal catalytic domain), directly confirming that PP_2325 possesses both the cofactor-binding and the catalytic domains characteristic of KPR.

The gene occupies genome coordinates 2,652,614–2,653,561 (948 bp, forward strand). Genomic-neighborhood analysis shows that PP_2325 is **not** part of a pantothenate operon: the upstream gene PP_2324 (K01626, DAHP synthase, aromatic amino-acid biosynthesis) is divergently transcribed on the opposite strand, and the downstream same-strand gene PP_2326 is an unrelated universal stress protein. The panB/panC/panD genes are located elsewhere in the genome. This gene arrangement is consistent with independent transcriptional regulation of ketopantoate reductase rather than co-regulation within a biosynthetic gene cluster.

Finding 8 — *P. putida* encodes a complete pantothenate pathway with a panBC cluster, an IlvC backup, and three KPR-family paralogs; PP_2325 is the canonical curated PanE/ApbA

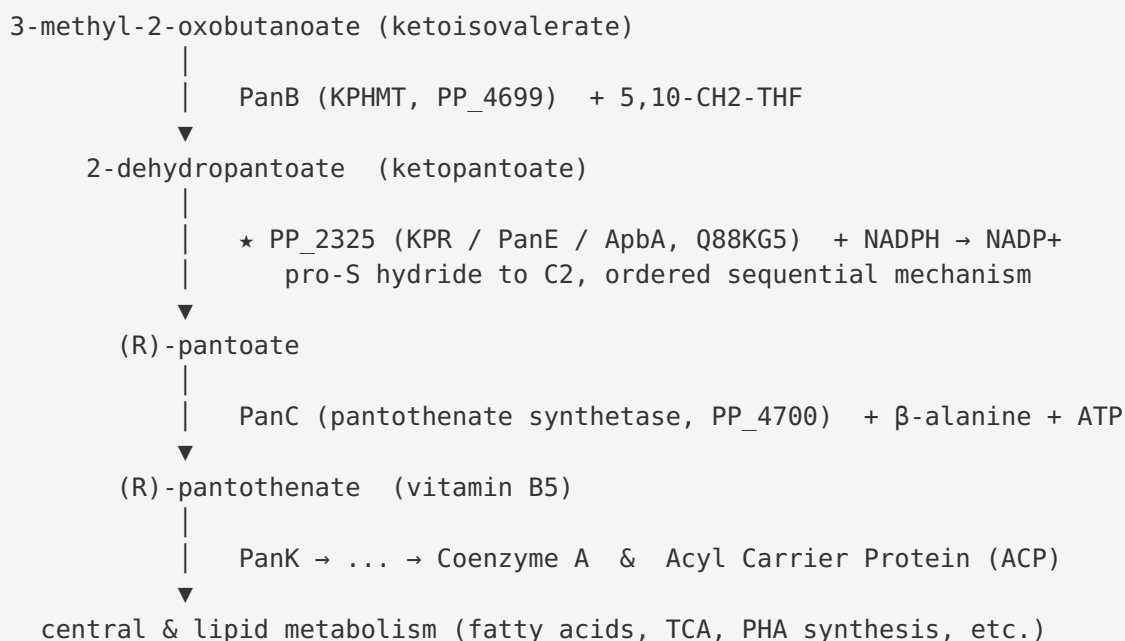
KEGG KO-to-gene mapping for *P. putida* KT2440 (ppu) resolves the pathway components: panB = K00606 → PP_4699 and panC = K01918 → PP_4700 (adjacent loci forming a panBC cluster, located far from PP_2325); ilvC = K00053 → PP_4678 (present, providing the documented secondary/moonlighting ketopantoate reductase activity). Notably, three genes are assigned to K00077 (2-dehydropantoate 2-reductase): **PP_1351, PP_2325, and PP_2998** — indicating multiple KPR-family paralogs in this genome, consistent with the "KPR-Related" InterPro superfamily IPR051402. PP_2325 is the paralog carrying the canonical ApbA/PanE UniProt curation (Q88KG5) and both ApbA + ApbA_C Pfam domains.

KEGG returned no gene under panD (K01579, aspartate 1-decarboxylase), suggesting that the β -alanine supply for pantothenate in *P. putida* may proceed via an alternative or divergent route. The existence of an IlvC ortholog plus two additional K00077 paralogs implies that a single PP_2325 knockout might be phenotypically buffered, mirroring the *E. coli* situation where an ilvC panE double mutant is required to fully abolish KPR activity.

Mechanistic Model and Interpretation

The reaction and its place in metabolism

PP_2325 catalyzes the second (committed, reductive) step of the pantoate branch of pantothenate biosynthesis:



The enzyme is the sole dedicated reductase of this branch and produces the chiral (R)-pantoate needed downstream. Because CoA and ACP are indispensable for fatty-acid metabolism, the TCA cycle, and — importantly for *P. putida* — medium-chain-length polyhydroxyalkanoate (PHA) synthesis and diverse aromatic/terpenoid catabolism, KPR sits upstream of a large fraction of the organism's carbon flux.

Enzyme architecture and catalytic cycle

The structural and kinetic model, transferred from characterized bacterial orthologs, is:

Feature	Property (from <i>E. coli</i> PanE)	Basis for transfer to PP_2325
Oligomeric state	Monomer, ~34 kDa	PP_2325 = 315 aa, ~34 kDa
Fold	Two domains: N-terminal Rossmann + C-terminal α -helical	ApbA (PF02558) + ApbA_C (PF08546) present
Cofactor	NADPH (binds first)	GSGAIG Rossmann motif, res 12–17
Substrate	Ketopantoate (binds second)	Family / KO assignment K00077
Kinetic mechanism	Ordered sequential; pantoate released before NADP ⁺	Family-conserved
Chemistry	Stereospecific pro-S hydride transfer to C2 $\rightarrow$ (R)-pantoate	Family-conserved
Catalytic residues	Asn98, Lys176, Glu256 (catalysis); Arg31, Lys72 (cofactor); Ser244 (substrate)	Encoded in InterPro HMM; not verifiable by naive alignment
Conformational change	Hinge-bending domain closure repositions Lys176	Family-conserved

An important nuance emerges from comparative studies: while *E. coli* KPR is monomeric and follows a strictly ordered mechanism, the *Staphylococcus aureus* enzyme is a stable dimer showing kinetic cooperativity for NADPH (Hill coefficient ~2.5) that arises from a random-addition mechanism kinetically biased toward NADPH-first binding [PMID: 25946571](#). The oligomeric state and exact kinetic behavior of PP_2325 therefore cannot be assumed a priori and would require direct biochemical study.

Localization

Ketopantoate reductase is a soluble cytoplasmic enzyme. It has no signal peptide, no transmembrane segments, and its substrates/products (ketopantoate, pantoate, NADPH/NADP⁺) are cytosolic metabolites. Its role is carried out entirely within the cytoplasm, where it participates in soluble intermediary metabolism.

Redundancy and regulation

The *P. putida* genome presents a more complex picture than a single dedicated KPR. Three genes map to K00077 (PP_1351, PP_2325, PP_2998), and an IlvC ortholog (PP_4678) can supply moonlighting KPR activity — a phenomenon experimentally demonstrated in other systems, where *"functional replacement of a pantothenate biosynthesis enzyme, 2-dehydropantoate 2-reductase (E.C. 1.1.1.169), by an endosymbiont gene, ilvC, encoding a substrate ambiguous enzyme"* was shown [PMID: 25527092](#). PP_2325 is nonetheless the canonically curated PanE/ApbA reductase, distinguished by its UniProt curation and complete ApbA + ApbA_C domain complement. Its non-operonic genomic context (isolated from panBC/panD) implies independent regulation.

In archaea, KPR is a feedback regulatory node for CoA biosynthesis (CoA competitively inhibits at the NAD(P)H site) [PMID: 23941541](#); [PMID: 26757028](#). Whether the *P. putida* enzyme is similarly feedback-regulated is unknown, as bacterial CoA feedback regulation classically targets pantothenate kinase rather than KPR.

Evidence Base

PMID	Title (abbreviated)	How it supports the annotation
15966718	<i>Crystal structure of E. coli KPR with NADP⁺ bound</i>	Establishes reaction, NADPH cofactor, EC 1.1.1.169, pathway role, and two-domain cofactor binding
17229734	<i>E. coli KPR ternary complex with NADP⁺ and pantoate</i>	Identifies catalytic residues (Asn98, Lys176, Glu256), substrate recognition, conformational change
10736170	<i>Kinetic/mechanistic analysis of E. coli panE KPR</i>	Ordered sequential mechanism; stereospecific pro-S hydride transfer to C2
11724562	<i>E. coli KPR crystal structure at 1.7 Å</i>	Monomeric two-domain Rossmann architecture; 6-PGDH C-terminal superfamily assignment
18726075	<i>Engineering pantothenate levels in plants</i>	Defines the four-enzyme bacterial pathway (panB/panD/panC/panE) placing KPR as the reductase
9721324	<i>panE maps at 10 min, allelic to apbA in S. typhimurium</i>	Establishes ApbA = PanE identity
9488683	<i>ApbA of S. typhimurium required for thiamine via APB pathway</i>	Confirms ApbA is ketopantoate reductase; monomer, NADPH-preferring
20876192	<i>Detecting subtle functional differences in KPR</i>	Documents limited KPR sequence similarity; justifies HMM-based annotation
20158506	<i>Conditionally essential genes of P. putida KT2440</i>	Organism-specific evidence that vitamin biosynthesis genes are conditionally essential on minimal medium
25946571	<i>Kinetic cooperativity in dimeric S. aureus KPR</i>	Shows oligomeric state/kinetics vary across KPR family — cautions against over-transfer
25527092	<i>Substrate-ambiguous enzyme in symbiont genome reduction</i>	Demonstrates IlvC can functionally replace KPR (EC 1.1.1.169), supporting redundancy model
23941541	<i>Archaeal KPR and CoA biosynthesis regulation</i>	KPR as a feedback regulatory node in archaea; NADH-preferring variant exists
26757028	<i>Archaeal KPR complexed with CoA and 2-oxopantoate</i>	Structural basis for CoA feedback inhibition at the NAD(P)H site
23243306	<i>PanG, a new ketopantoate reductase</i>	Shows multiple non-orthologous KPRs exist; convertibility of KPRs to pantoate

PMID	Title (abbreviated)	How it supports the annotation
10553653	<i>Pantothenate production via panE overexpression</i>	panE encodes 303-aa, 33.8 kDa KPR; overexpression boosts pantothenate excretion

The **consistency across UniProt, KEGG, InterPro/Pfam, and the primary structural/kinetic literature** is the strongest aspect of this annotation: four independent annotation systems and multiple experimentally characterized orthologs converge on the same assignment. The primary **challenge** is the absence of direct experimental characterization of the *P. putida* PP_2325 protein itself — all mechanistic detail is transferred from orthologs — combined with the family's low sequence identity and evidence that oligomeric state/kinetics vary between homologs.

Limitations and Knowledge Gaps

- 1. No direct experimental characterization of PP_2325.** There is no published enzymology, crystal structure, or knockout phenotype for the *P. putida* KT2440 protein specifically. All mechanistic claims (kinetics, stereochemistry, catalytic residues, oligomeric state) are inferred from orthologs, principally *E. coli* PanE.
- 2. Low sequence identity limits residue-level inference.** At ~27% identity to *E. coli* PanE, naive pairwise alignment cannot register the catalytic residues reliably (Finding 6). Residue equivalence rests on profile-HMM assignment. A structure-based alignment (e.g., against an AlphaFold model) was not performed and would strengthen residue-level claims.
- 3. Family-level variability in mechanism.** KPRs range from monomeric/ordered (*E. coli*) to dimeric/cooperative (*S. aureus*), and some prefer NADH over NADPH (archaeal Tk-KPR). The oligomeric state, cofactor preference, and kinetic mechanism of PP_2325 are therefore not settled by homology alone.
- 4. Genetic redundancy complicates loss-of-function interpretation.** With three K00077 paralogs (PP_1351, PP_2325, PP_2998) plus a potentially moonlighting IlvC (PP_4678), a single PP_2325 deletion may not produce a pantothenate auxotrophy, obscuring in vivo confirmation of its role.
- 5. Missing/divergent panD.** No canonical aspartate 1-decarboxylase (K01579) was mapped, leaving the β -alanine supply route for pantothenate synthesis in *P. putida* unresolved. This is a pathway-level gap, not specific to PP_2325.

6. **Regulation unknown.** Whether PP_2325 is feedback-inhibited by CoA (as in archaea) or transcriptionally regulated (its non-operonic context suggests independent control) has not been investigated.

Proposed Follow-up Experiments and Actions

1. **Recombinant expression and enzyme assay.** Clone PP_2325, purify the protein, and measure NADPH-dependent reduction of chemically synthesized ketopantoate spectrophotometrically (NADPH oxidation at 340 nm). Determine $K_M(\text{NADPH})$, $K_M(\text{ketopantoate})$, and k_{cat} , and test NADH vs NADPH preference to place PP_2325 relative to bacterial vs archaeal KPR types.
2. **Determine oligomeric state.** Use size-exclusion chromatography and/or analytical ultracentrifugation to distinguish a monomeric (*E. coli*-like) from a dimeric (*S. aureus*-like) assembly, and test for kinetic cooperativity via NADPH saturation curves (Hill coefficient).
3. **Structural validation.** Generate an AlphaFold model of Q88KG5 and perform a structure-based superposition against *E. coli* KPR (structures from

 11724562

 17229734) to confirm spatial conservation of the Asn98/Lys176/Glu256-equivalent residues; ideally solve an experimental structure with NADP⁺ and pantoate bound.
4. **Genetic dissection of redundancy.** Construct single (ΔPP_{2325}) and combinatorial ($\Delta\text{PP}_{2325} \Delta\text{PP}_{1351} \Delta\text{PP}_{2998} \Delta\text{ilvC}$) deletion mutants and score for pantothenate auxotrophy on minimal medium $\pm$ pantothenate/pantoate to establish which paralog(s) carry the physiological KPR activity in vivo.
5. **Complementation test.** Confirm functional identity by complementing an *E. coli* *ilvC* *panE* double mutant (which lacks all KPR activity) with PP_2325, as was done successfully for *panG* and EF1861 [PMID: 23243306](#).
6. **Feedback-regulation test.** Assay PP_2325 activity in the presence of CoA to determine whether this bacterial enzyme, like its archaeal counterparts, is subject to CoA feedback inhibition at the nucleotide-binding site.
7. **Resolve the β -alanine route.** Search the *P. putida* genome/transcriptome for a divergent aspartate decarboxylase or an alternative β -alanine source (e.g., from spermine/uracil degradation) to complete the pathway annotation.

Conclusion

PP_2325 (Q88KG5) is the ketopantoate reductase (2-dehydropantoate 2-reductase; PanE/ApbA; EC 1.1.1.169; K00077) of *Pseudomonas putida* KT2440 — a soluble cytoplasmic, NADPH-dependent oxidoreductase that stereospecifically reduces ketopantoate to (R)-pantoate in the committed reductive step of pantothenate (vitamin B5) biosynthesis, upstream of coenzyme A and acyl carrier protein. This assignment is supported by convergent UniProt, KEGG, and InterPro/Pfam annotations, by the diagnostic Rossmann NADP-binding motif and dual ApbA/ApbA_C domains, and by extensive biochemical and structural characterization of bacterial orthologs, with the caveats that the family is highly divergent, that *P. putida* harbors partially redundant KPR-family paralogs and a moonlighting IlvC, and that no direct experimental characterization of the PP_2325 protein itself yet exists.

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